

Slicing Energy Use with Sharp Calculation Tools



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How can a calculator save energy? The right kind of calculator can provide a rational basis for deciding which conservation measures are good investments for a particular building. Builders and consumers alike need a credible source of unbiased, reliable information, so Oak Ridge National Laboratory (ORNL) has developed a collection of Web-based energy-efficiency calculators for buildings (www.ornl.gov/roofs+walls).

Roof Radiation Control Calculator

The Roof Radiation Control Calculator estimates the annual cost savings from installing radiation control on a low-slope roof. The calculator uses several variables. Solar reflectance determines how much solar energy the roof will reflect, and infrared emittance determines how much thermal energy is radiated away from a building. Infrared emittance is also important at night, because a roof with a greater infrared emittance will release more heat to the cold night sky, which is desirable in the summer but not in the winter.

Another key factor in estimating radiation control is the amount of insulation beneath the roof surface. The more insulation there is, the hotter the roof surface will be during the day, which in turn will increase the amount of absorbed solar energy given off by infrared radiation and carried away by the wind. Further, tests have shown that the radiation properties

of roof surfaces change over time as the roof becomes dirty. ORNL's Buildings Technology Center has tested a wide range of reflective roof surfaces over three-year periods. Even if the surfaces are washed periodically, the radiation properties never quite return to their 'good-as-new' values. (ORNL's *Radiation Control Fact Sheet* gives more details.)

An hourly computer model that considered all these factors was validated with test data and then run over a number of climates for various roofs. The climates ranged from mostly cooling to mostly heating, and included Phoenix, Miami, Tampa, Dallas, Knoxville, Boulder, Minneapolis, and Anchorage. The low-slope roofs considered had lightweight decks and insulation levels ranging from R-5 to R-32. Seven combinations of solar spectrum reflectance and infrared emittance were modeled. They corresponded to values measured after two years in a comprehensive study of the thermal performance of 24 different coatings and four uncoated specimens, including white latex coatings, aluminum coatings, an asphalt emulsion, an aluminum capsheet, and an uncoated roof membrane.

An estimating tool was developed by correlating the results from these hourly simulations to the solar reflectance and infrared emittance of the roof surface, roof thermal resistance, cooling and heating degree-days, and average daily solar radiation. The calculator can use this correlation for the user's choice of location to calculate the change in summer cooling energy use and winter heating energy use (see "Radiation Control in Fort Worth"). Using entered heating and cooling system efficiencies and utility rates, the calculator will then show changes in annual utility costs if one

installs a radiation control roof surface. The calculator provides one more piece of information; it lets one compare installing a solar control roof surface to changing the amount of roof insulation, by indicating how much insulation one would have to add to achieve the same results. An additional capability is being developed to estimate hourly peak cooling savings for the summer months.

Insulation Tool

The Insulation ZIP-Code Calculator provides advice for adding insulation to a new or existing house. It was developed as a part of the *DOE Insulation Fact Sheet*, which is also available on the Web. This calculator was used to generate the official DOE insulation R-value recommendations, which are cited by many manufacturers and conservation organizations.

In addition to zip code, which is used to assign weather factors and cost multipliers, users enter heating and cooling system types and efficiencies. Users may enter their own cost data for insulation products and utility prices, or they can use the default values provided. These utility and fuel price default values are based on statewide average prices. The insulation cost default values are based on a national survey and state-specific insulation cost multipliers. These data are then combined in a life cycle cost optimization calculation to determine the appropriate amount of insulation for each part of a house, including the attic, cathedral ceilings, wall cavities, wall sheathing, floors, crawlspace, foundation perimeters, and basement walls. The calculator gives separate recommendations depending upon whether the home is wood framed, metal framed, or masonry.

Radiation Control in Fort Worth

A calculation of the energy saved by applying radiation control to a roof in Fort Worth, Texas, would yield a net annual savings of about a dime for each square foot of roof area (see Table A).

The input values include: an air conditioning coefficient of performance of 2, a heating system efficiency of 0.7, an electricity cost of

\$0.10/kWh, a fuel cost of \$1/therm, and a proposed roof R-value of 10. The proposed roof solar reflectance is 0.70 and the proposed roof emissivity is 0.90, representing a high reflectance white latex coating. For this example, the building operator could achieve the same savings with a conventional roof by increasing the roof system's R-value from 10 to 13.7.

Table A. Applying Radiation Control

	Load (Btu/ft ² -year)	Savings (\$/ft ² -year)
Cooling	7,935	0.116
Heating	-1,306	-0.019
Net		0.098

For example, for a new wood-framed house in Pittsburgh with a high efficiency gas furnace and a medium efficiency central air conditioner, the Zip-Code Calculator recommends attic insulation of R-49, wall cavity insulation of R-13 with an R-5 insulative wall sheathing, and floor insulation of R-25. If the Pittsburgh house were an existing home, rather than new construction, the attic recommendation would be R-38, the wall cavity value would be R-11, the wall sheathing would be R-7 (but only for cases where the exterior siding was removed for other remodeling purposes), and the floor level would be R-19. The values differ because the life cycle calculation assumes that the insulation added to a new home will have a longer life than in an existing home, and because some insulation measures are less costly in a new home than in a retrofit situation.

Whole-Wall Tool

The Whole-Wall Calculator (see “Calculating Whole Wall R-Values on the Net,” *HE* Nov/Dec '99, p. 22) can be used to make an informed decision about the energy performance of different wall types when building or remodeling. Often wall R-values are given in terms of the amount of insulation between the studs, which ignores the higher rate of heat transmission through the studs themselves. Other times, wall R-values are given in terms of a “clear wall,” which means that the overall R-value has been weighted to include the effect of the framing, but only for a plain wall without architectural details. But real wood- and steel-framed walls have much more framing than the 2 x 4s included in the clear-wall definition. They have additional framing members at corners where two walls meet, where the wall meets the floor and the roof, and to add strength at window and door openings.

Not only do these architectural details tend to transfer heat faster than the surrounding wall materials, but they also change the temperature distribution within the wall, and therefore influence the heat transfer through a much greater portion of the wall than just the area they cover. Indeed, for complex building designs, the thermal zone of influence of the wall details may exceed 30% of the total opaque wall area. And the heat

losses generated by these wall details can exceed 50% of the whole wall's heat losses—losses that would be missed in a clear-wall rating. This is why it is important to use a whole-wall rating system when comparing different types of wall. Understanding the effects of the complex heat transfer paths through this additional framing is essential if one is comparing walls made from different materials, such as wood framing versus metal framing. It is also critical if one is comparing framed walls to other types of wall, such as structural insulated panels or filled concrete forms.

A computer model has been used to consider many different combinations of house shapes and window and door placements. This computer model has been tuned to match experimental data from more than 100 wall test sections that were constructed to include the additional architectural features found in a real house. The results of this extensive analysis are available to builders and house planners in a convenient tool that lets the user select from several building types and window and door arrangements. The user can also specify all the layers within the planned wall. The resulting whole-wall R-value can then be used to compare the energy performance of the different types of wall under consideration. Look for more updates to the tool by year's end.



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For more information:

Radiation Control Fact Sheet URL:
www.ornl.gov/roofs+walls/facts/RadiationControl.htm

DOE Insulation Fact Sheet URL:
http://www.ornl.gov/roofs+walls/insulation/ins_01.html

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